

Resonance Quantum Theory (RQT): A Framework for Resonance-Base Structure and Interaction Conceptual Foundations and the Roton Quantum Model

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Abstract

The Resonance Quantum Theory (RQT) is a conceptual framework that interprets physical structure, interaction, and persistence as consequences of resonance organization rather than as properties imposed through fundamental particles, fields, or quantization rules. The framework begins from the assumption that localized resonance relationships can form within a continuous resonance-supporting continuum, giving rise to stable structures, interaction channels, and hierarchical organization across multiple scales.

In RQT, observable physical entities are viewed as dynamically maintained resonance structures. Stability arises through resonance closure, while interactions emerge through open and residual resonance channels connecting otherwise localized systems. Information, inertia, and binding are interpreted as consequences of the accommodation and reconfiguration of these resonance relationships. Matter and radiation are therefore treated not as fundamentally distinct categories, but as different manifestations of resonance organization within a common underlying framework.

The theory further proposes that physical structure develops hierarchically. Larger and more persistent systems emerge from networks of mutually stabilizing resonance configurations, while observable forces reflect residual effects of incomplete resonance closure. In this view, quantization, stability, and long-range interaction arise as natural consequences of resonance geometry and organization rather than as independent postulates.

To facilitate visualization, analysis, and numerical investigation, the Roton Quantum Model (RQM) is introduced as a discrete structural representation of the broader RQT framework. RQM projects resonance structures onto simplified resonance centers and channels, allowing complex resonance relationships to be described through a tractable geometric model while preserving the central concepts of closure, hierarchy, and interaction.

The purpose of this paper is to establish the conceptual foundations of RQT, define its principal structural concepts, introduce the relationship between RQT and RQM, and explore how resonance-based organization may provide a unified description of physical structure across scales.

1 Introduction

Nature exhibits persistent structure across an extraordinary range of scales. Atomic systems form stable molecules, molecules organize into complex materials, and larger structures emerge through increasingly intricate patterns of interaction. Despite their diversity, such systems share a common characteristic: they maintain coherence, stability, and recognizable identity while continuously interacting with their surroundings.

Modern physics provides highly successful mathematical descriptions of these phenomena. Quantum theory, quantum field theory, and relativity accurately predict a vast range of experimental observations and have established a remarkably consistent framework for understanding physical behavior. At the same time, many fundamental properties of physical systems are introduced through independent postulates, parameters, symmetries, or interaction rules. Stability,

quantization, binding, charge, and long-range interaction are typically described within specialized theoretical frameworks, each optimized for a particular domain of observation.

This raises a broader conceptual question: might these apparently distinct phenomena arise from a common underlying organizational principle? Rather than beginning with particles, fields, or forces as fundamental entities, one may instead ask whether physical structure itself emerges from the formation and persistence of relationships between interacting systems.

The Resonance Quantum Theory (RQT) explores this possibility. The framework is based on the assumption that resonance relationships can form between localized structures and that these relationships may themselves become stable, persistent, and capable of further interaction. In this view, physical systems are understood not primarily as isolated objects but as dynamically maintained networks of resonance relationships. Stability, interaction, and hierarchy are interpreted as consequences of the organization of these relationships across different scales.

The purpose of RQT is not to replace successful physical descriptions, but to investigate whether a common resonance-based framework can provide a deeper conceptual basis from which a wide range of observed phenomena may emerge. The theory therefore focuses on the formation of resonance structures, the development of stable interaction channels, and the mechanisms through which increasingly complex systems can arise from comparatively simple assumptions.

To facilitate visualization and quantitative exploration of these concepts, a simplified structural representation referred to as the Roton Quantum Model (RQM) is introduced later in this work. RQM provides a discrete geometric description of resonance structures and their interactions while remaining rooted in the broader conceptual framework established by RQT.

The following chapters develop the fundamental concepts of the theory, introduce the notion of resonance channels and closure, and examine how physical structure may emerge from the organization of resonance relationships across multiple levels of complexity.

2 Conceptual Foundations

The conceptual foundation of RQT is intentionally minimal. Rather than beginning with particles, fields, or forces, it starts from the possibility that stable identities may emerge through persistent relationships. Once such an identity has formed, it may itself participate in further relationships, allowing increasingly complex structures to arise through repeated application of the same principle. Figure 1 summarizes this recursive concept, which is developed throughout the remainder of this section.

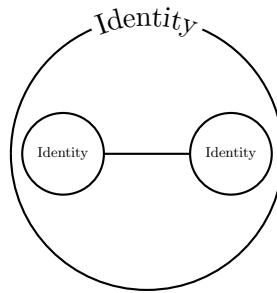


Figure 1: Recursive Formation of Identity

The remainder of this section develops the individual conceptual elements of this recursive process.

2.1 Persistence and Self-Resonance

Before a structure can interact, it must first persist.

The Resonance Quantum Theory begins with a minimal conceptual assumption: the existence of a continuum capable of supporting oscillatory behavior. The framework does not initially specify the origin, composition, or geometry of this continuum. It merely assumes that localized oscillatory configurations may arise within it.

An oscillatory configuration does not necessarily constitute a persistent structure. Of particular interest are configurations that repeatedly return to compatible states. Such recurrence allows an oscillation to maintain its identity over extended durations and establishes the possibility of persistence.

Within RQT, persistence is regarded as the first prerequisite for the emergence of structure. A persistent oscillatory configuration forms a self-maintaining cycle whose future evolution remains compatible with its preceding states. This process is referred to as self-resonance. At the conceptual level adopted here, self-resonance may be regarded as the simplest form of persistent resonance behavior.

The precise nature of such self-resonant configurations is not specified at this stage. Their significance lies not in their detailed structure, but in the fact that they provide a mechanism through which stable and persistent patterns may arise. Once persistence is established, increasingly complex forms of coupling, interaction, and organization become possible.

RQT therefore considers persistent self-resonant configurations as the conceptual starting point from which higher-order resonance structures can emerge.

2.2 Stability and Identity

Persistence allows oscillatory behavior to remain. Stability allows it to become structure.

A persistent oscillatory configuration does not necessarily remain unchanged. Local adjustments, interactions, and ongoing accommodation may continuously influence its evolution. Within RQT, stability therefore does not imply rigidity or immutability. Instead, a structure is considered stable when it repeatedly returns to sufficiently compatible states, allowing its characteristic behavior to persist despite ongoing change.

Such stable resonance structures represent the first distinguishable configurations within the framework. Their identity does not arise from a fixed internal composition, but from the persistence of recognizable resonance properties over time. A stable structure may therefore be characterized through its phase relationships, recurrence behavior, and interaction potential rather than through a permanently defined geometry.

The continued existence of a stable resonance structure also preserves information about its formation and evolution. This information is not regarded as a separately stored quantity, but as an intrinsic consequence of the ongoing resonance process itself. The present state of a structure remains connected to preceding states through a continuous resonance flow, allowing distinctions and relationships to persist over time.

Once stable resonance structures emerge, they may participate in increasingly complex forms of coupling and organization. Stability therefore provides the foundation upon which interaction, relationship formation, and higher-order resonance structures can develop.

Within RQT, stable resonance structures constitute the first identifiable building blocks from which larger resonance networks and hierarchical organizations may arise.

2.3 Interaction and Relationship Formation

Interaction creates relationships. Relationships create shared identity.

Stable resonance structures do not exist in complete isolation. Whenever the state of one structure influences the evolution of another, an interaction is established. The precise mechanism through which this influence occurs is not specified at this stage. It is sufficient to recognize that persistent structures may affect one another and thereby participate in a common process of development.

Within RQT, interactions are significant not merely because they modify existing structures, but because they give rise to relationships. A relationship represents an organized connection between two or more structures whose future evolution can no longer be described entirely independently. Through interaction, participating structures begin to share aspects of their resonance behavior and thereby become part of a larger organizational whole.

The emergence of a relationship introduces an additional level of identity. While the participating structures may remain individually distinguishable, the relationship itself acquires significance as a persistent element of the overall resonance organization. From a higher-level perspective, the relationship may therefore be regarded as a structure in its own right.

Relationships need not possess equal strength or persistence. Some may remain weak and contribute only limited influence on the participating structures. Others may develop increasingly shared resonance organization, causing their behavior to become progressively interdependent. The distinction between weak, strong, and highly persistent relationships is therefore not regarded as fundamental, but as a consequence of the degree to which resonance organization becomes shared.

As relationships persist, they may themselves participate in further interactions and form new relationships with other structures. In this manner, interaction becomes a mechanism through which increasingly complex organizations can emerge. Relationship formation therefore provides the foundation for higher-order resonance structures and the hierarchical organization discussed in the following section.

2.4 Emergence and Hierarchical Organization

Every stable organization becomes a candidate for higher-order organization.

The formation of a relationship does not necessarily result in a new persistent identity. Many relationships may remain transient, while others persist only for limited durations. Within RQT, higher-order structures emerge when relationships acquire sufficient stability to participate in the resonance organization of their surroundings as identifiable entities in their own right.

The persistence of a relationship depends on the degree of compatibility between the participating structures. Compatibility is not regarded as a binary property, but as a condition that may develop through continued interaction. As structures influence one another, adjustments in their relative states may increase or decrease the stability of the relationship. Persistent relationships therefore arise not merely from interaction itself, but from the ability of interacting structures to accommodate one another sufficiently to maintain a shared organization over time.

When a relationship becomes persistent, a new level of identity emerges. This higher-order identity is not simply the sum of its constituents. Instead, it may exhibit resonance capabilities, interaction possibilities, and organizational properties that were not available to the participating structures independently. The emergence of such capabilities constitutes the basis of higher-order structure formation within the framework.

The formation of a new identity may also alter the accessibility of previously available relationships. Certain interactions may become enhanced, while others may become reduced, concealed, or effectively internalized within the newly formed structure. Emergence therefore introduces not only new possibilities for interaction, but also new forms of organization and selective accessibility.

Once established, higher-order identities may themselves participate in further relationship formation. The same principles that govern the emergence of lower-level structures may therefore operate recursively across multiple levels of organization. In this manner, increasingly complex resonance structures can arise without requiring additional foundational assumptions.

RQT consequently interprets hierarchy not as a separate organizing principle, but as a natural consequence of persistent relationship formation, emergent identity, and the continual development of new resonance capabilities across successive levels of organization.

2.5 Accessible and Enclosed Structure

Stability favors enclosure. Enclosure creates selective accessibility.

The emergence of higher-order structures introduces an important distinction between the internal organization of a structure and the properties it exposes to its surroundings. Within RQT, stable structures are not assumed to make all aspects of their internal resonance organization equally accessible. Instead, the formation of persistent identities naturally leads to varying degrees of accessibility between internal and external levels of organization.

This tendency arises from the continuous search for stable resonance configurations. As relationships become organized into higher-order structures, the participating components may reduce their dependence on external interactions by establishing increasingly stable internal relationships. The resulting organization can allow parts of the internal structure to become less directly involved in the structure's interactions with its surroundings.

Enclosure is therefore not regarded as a separate organizing principle, but as a natural consequence of stability. A structure that reduces unnecessary external dependencies may become less susceptible to disturbance and thereby maintain its identity more effectively over time. The degree of enclosure achieved by a structure depends on both its internal organization and the conditions imposed by its environment.

The emergence of enclosure creates selective accessibility. Some properties of a structure remain available for further relationship formation and interaction, while others become partially or fully internalized within the structure itself. As a result, the externally accessible behavior of a structure need not reveal the full complexity of its internal organization.

The accessible properties of a structure define how it participates in higher-order organization. Internal relationships, however, may continue to influence the behavior and stability of the structure even when they are not directly accessible from the outside. In this sense, every emergent structure may possess both an externally visible identity and an internal organization that extends beyond what is immediately available to its surroundings.

RQT therefore regards accessibility and enclosure as complementary aspects of stable organization. The formation of higher-order structures not only creates new identities and interaction possibilities, but also establishes boundaries between what remains externally available and what becomes incorporated into the internal organization of the structure.

2.6 Framework Assumptions and Scope

Complexity is not introduced. Complexity is accumulated.

The preceding sections have established the conceptual foundations upon which the Resonance Quantum Theory is developed. Rather than beginning with particles, forces, fields, or predefined physical laws, the framework starts from a small set of organizational assumptions and explores the consequences that follow from them.

RQT assumes the existence of a continuum capable of supporting oscillatory behavior. Within this continuum, persistent oscillatory configurations may arise and maintain their identity through self-resonance. Persistent structures may influence one another, form relationships, and develop increasingly stable forms of organization. Relationships that become sufficiently persistent may themselves emerge as higher-order identities capable of participating in further relationship formation.

The framework does not assume that all configurations persist. Persistence is itself regarded as a non-trivial property. Only structures that remain sufficiently stable can continue to participate in subsequent organization and relationship formation. Increasing levels of complexity may therefore arise through the continued accumulation of persistent relationships across multiple levels of organization.

From these assumptions emerge the concepts of identity, interaction, hierarchy, accessibility, and enclosure. Higher-order structures are understood not as independent organizing princi-

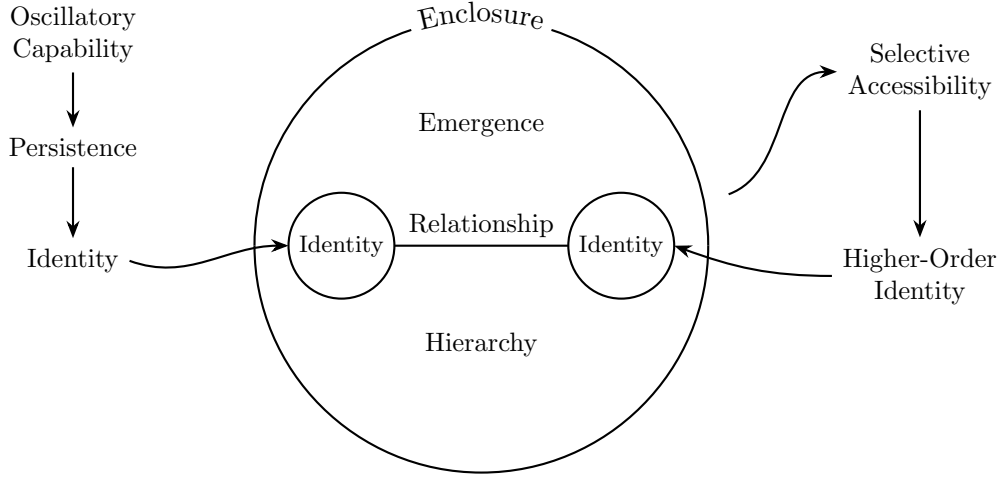


Figure 2: Conceptual foundations of the Resonance Quantum Theory

ples, but as consequences of recursive relationship formation and the development of persistent resonance organization.

At this stage, the framework intentionally refrains from specifying the detailed mechanisms through which such structures are realized. It does not initially assume particles, forces, charge, mass, spacetime geometry, probabilistic state descriptions, or specific interaction laws. These concepts, if required, are expected to emerge as effective descriptions of deeper resonance processes rather than as fundamental starting points.

The purpose of the framework is therefore not to introduce additional elementary entities, but to investigate how increasingly complex physical organization may arise from a minimal set of resonance-based assumptions. The following sections explore how specific resonance structures, interaction mechanisms, and observable physical phenomena may emerge from these principles.

3 The Resonance Quantum Theory

Conceptual foundations of the framework.

The conceptual foundations introduced the recursive emergence of persistent identities without assuming a specific physical realization. The Resonance Quantum Theory extends this framework by describing how such identities interact, organize, and evolve through resonance itself. The concepts introduced in this section remain independent of any particular geometric or discrete representation and therefore form the theoretical basis upon which the Roton Quantum Model is later constructed.

3.1 Resonance as a Fundamental Principle

Resonance is not treated in RQT as a special physical phenomenon occurring only under particular conditions. Instead, it is adopted as the fundamental principle by which oscillatory structures establish, maintain, and evolve stable organization.

The conceptual foundations introduced the emergence of persistent identities through recursive relationship formation without assuming a particular physical realization. The next step is to identify the principle that determines whether such relationships remain transient or develop into stable structures. RQT proposes that this principle is resonance.

In its most general sense, resonance describes the mutual adaptation of oscillatory systems whenever their evolving states become sufficiently compatible. Such adaptation may involve phase, characteristic frequencies, spatial organization, or other structural properties, without

requiring any specific geometric representation. While many interactions remain temporary, resonance provides the possibility for oscillatory systems to organize into persistent configurations.

From the perspective of RQT, resonance is therefore not regarded as a consequence of existing physical objects. Rather, stable physical objects are understood as consequences of persistent resonance. Resonance is treated as the organizing principle from which increasingly stable structures may emerge, interact, and recursively form higher-order identities.

The following sections develop this framework by introducing the continuum in which resonance occurs, the formation of localized resonance structures, and the mechanisms by which resonance accommodates, stabilizes, and eventually gives rise to observable physical systems.

3.2 The Resonance Continuum

RQT does not begin by asking what space is made of. It begins by asking what is minimally required for stable resonance to exist.

RQT does not attempt to explain the origin or fundamental nature of the substrate in which resonance occurs. Instead, it adopts the minimal assumption that a continuum exists which is capable of supporting persistent oscillatory organization. Beyond this requirement, no particular physical interpretation is assumed.

From the perspective of RQT, this continuum should not initially be identified with physical space or spacetime. Rather, it represents the possibility for oscillatory structures to emerge, persist, and mutually influence one another. Observable notions such as separation, spatial distance, and temporal evolution are not taken as fundamental prerequisites but are considered possible emergent descriptions of stable resonance organization.

This represents a conceptual inversion of the usual viewpoint. Rather than assuming that resonance occurs within a pre-existing spacetime, RQT explores the possibility that the observable notions of space and time arise from the formation, separation, and continued interaction of persistent resonance identities. The theory therefore begins with distinction and relationship rather than with metric geometry.

Only one assumption is required: sufficiently compatible oscillatory structures may establish persistent relationships. These relationships provide the basis from which increasingly organized resonance structures may emerge and subsequently interact through newly available resonance pathways. The concepts of distance, propagation, and geometry are introduced only insofar as they become meaningful descriptions of these emerging relationships.

A consequence of this viewpoint is that the underlying mechanism giving rise to the first persistent oscillatory capability need not itself remain directly accessible. If higher-order resonance structures arise through successive stages of resonance organization, then the deeper levels from which these structures originate may become intrinsically concealed by the same principles of resonance confinement that later govern stable physical systems. In this sense, the apparent simplicity of a persistent oscillation may itself be the observable residue of a deeper resonance organization whose internal structure no longer participates directly in accessible interactions.

The following sections therefore take the existence of this resonance continuum as a conceptual starting point and investigate how localized resonance structures emerge, interact, accommodate one another, and eventually give rise to stable physical organization.

3.3 Distinct Resonance Structures

The first observable property of a resonance structure is not its location, but its persistent distinction.

The existence of a resonance continuum alone does not imply the existence of distinguishable structures. Many oscillatory configurations may remain transient, continuously emerging and dissolving without establishing persistent organization. RQT therefore distinguishes between

arbitrary oscillation and oscillatory organization that becomes sufficiently persistent to form a distinct resonance structure.

A distinct resonance structure is not initially defined by position, geometry, dimensionality, or physical extent. Instead, it is defined by the continued persistence of its oscillatory organization and by its distinguishability from the surrounding resonance continuum. Distinction therefore precedes localization. Only after stable relationships emerge may geometric descriptions become meaningful.

Persistent distinction also implies continuity. A resonance structure possesses both a history and the potential for continued evolution, not because information is explicitly stored, but because its oscillatory organization remains continuously connected to its own past. The present state is therefore understood as the ongoing evolution of the same resonance process rather than as a sequence of independently recreated states.

From the perspective of RQT, a resonance structure remains part of the observable resonance organization only insofar as this continuity persists. A structure that became completely disconnected from all continuing resonance relationships would cease to participate in the same observable organization. Whether such complete decoupling is possible remains an open question. Everyday physical structures, however, exhibit persistent relationships over extended periods, suggesting that stable resonance organization generally preserves continuity rather than abandoning it.

RQT begins not with localized objects, but with persistent distinction. Identity emerges from the continued existence of a persistent resonance organization. Interaction, separation, and geometry emerge only thereafter through the relationships established between such structures.

Distinct resonance structures therefore provide the first entities capable of participating in persistent interaction. The following section examines how these structures influence one another, adapt through resonance, and establish the relationships from which increasingly organized physical systems emerge.

3.4 Resonance Interaction

Interaction begins whenever the continued evolution of one distinct resonance structure influences that of another.

The existence of distinct resonance structures inevitably allows the possibility of interaction. Whenever two structures participate in one another's continued evolution, they no longer develop independently. Their oscillatory organization begins to adjust through mutual influence, giving rise to resonance interaction.

Interaction should not be understood as a discrete event but as a continuous process. Some interactions remain transient, producing only temporary adjustments before the participating structures continue independently. Others persist, allowing the interacting structures to accommodate one another over extended periods. Resonance interaction therefore represents a dynamic process whose consequences depend on the continued compatibility of the participating structures rather than on the mere occurrence of a single encounter.

Accommodation does not necessarily imply observable change. Highly compatible resonance structures may remain in continuous interaction while exhibiting little or no apparent evolution because their oscillatory organization is already mutually consistent. Interaction therefore exists whenever resonance structures contribute to one another's continued organization, regardless of whether this contribution manifests as rapid adaptation or as stable equilibrium.

The degree to which an interaction contributes to the continued organization of the participating structures determines its significance within the evolving resonance system. Persistent interactions gradually become part of the resonance organization itself, while transient interactions leave little lasting influence. The following sections investigate how such persistent interactions organize into stable resonance relationships and eventually give rise to the structured resonance pathways that characterize stable physical systems.

3.5 Resonance Accommodation

Interaction becomes organization when resonance structures begin to accommodate one another.

Resonance interaction is not a fixed relation between otherwise unchanged structures. Continued interaction influences the evolution of all participating resonance structures and may gradually alter the compatibility of their oscillatory organization. RQT refers to this process as resonance accommodation.

Accommodation is understood as a continuous and mutual process. The participating structures do not merely act upon one another from the outside; their evolving resonance organization becomes increasingly interdependent. This may involve adjustments in phase, characteristic periods, relative organization, or other resonance properties, without requiring any particular geometric representation at this stage.

Successful accommodation allows interacting structures to develop a partially shared resonance evolution. Their future states can no longer be described independently, because each structure contributes to the continued organization of the other. Where compatibility increases, the interaction may persist and become part of a larger resonance organization. Where compatibility cannot be maintained, the interaction may remain transient, weaken, or lead to reorganization.

Accommodation does not necessarily produce visible change. Two highly compatible structures may remain in continuous interaction while exhibiting little apparent evolution, precisely because their resonance organization is already mutually consistent. Stable resonance therefore represents not the absence of interaction, but a condition in which further accommodation is only minimally required.

In realistic systems, accommodation is not limited to isolated pairs. Every resonance structure may participate in several simultaneous interactions, each contributing to its continued evolution. Resonance accommodation therefore occurs within a network of mutually constraining relationships, where the persistence of any one interaction depends on its compatibility with the wider resonance organization.

RQT does not require that resonance structures consciously or globally optimize their organization. It proposes only that interactions capable of maintaining compatible shared evolution remain available for continued organization, while incompatible interactions tend to remain transient or reorganize. In this sense, accommodation does not create resonance; it reveals which resonance relationships can persist.

The persistence of accommodated interaction provides the conceptual basis for the structured resonance relationships introduced in the following sections.

3.6 Stability and Persistence

Stability denotes the persistence of an organized resonance process, not the absence of change.

A resonance organization becomes stable when its continued evolution preserves the distinction and internal coherence by which it can be identified. Such stability does not require every oscillatory state or resonance relationship to remain unchanged. Rather, the organization must retain sufficient continuity for its present evolution to remain connected to its preceding states.

Even a simple resonance relationship contains both internal and external aspects. Internally, the participating structures sustain the mutual organization through which they form a distinguishable system. Externally, each participating structure may remain exposed to additional resonance influences arising from its surrounding environment or from relationships at another level of organization. Stability therefore depends not only on internal compatibility, but also on the capacity of the system to accommodate changing external demands.

External or internal changes may introduce resonance mismatch into an existing organization. A stable system can respond by adjusting and redistributing this mismatch across its participating relationships while preserving the organization as a whole. The resulting changes may occur

within subordinate structures, within the relationships that hold them together, or in the system's external relations, without necessarily interrupting its persistent identity.

Stability is therefore dynamic rather than static. A stable resonance organization may undergo continuous internal adjustment while remaining distinguishable as the same continuing process. The more effectively such adjustments remain bounded within the existing organization, the less they disturb its higher-level identity.

Persistence and stability describe related but distinct aspects of this continuation. Persistence refers to the continued existence of a distinguishable resonance organization. Stability refers to its capacity to preserve that continued identity under changing resonance conditions.

When an imposed resonance mismatch can no longer be accommodated within the existing organization, the system may reorganize, separate, or dissolve into configurations capable of more persistent resonance evolution. Separation is therefore not necessarily the failure of accommodation, but may itself restore compatibility among the remaining resonance structures.

A stable resonance system is thus not isolated from change. Its stability lies in maintaining a recognizable organization while continuously accommodating the resonance influences in which it participates.

3.7 Resonance Distribution Across Hierarchies

Resonance organization is sustained not only by local accommodation, but by the distribution of accommodation demands across nested structures and relationships.

A distinguishable resonance structure may itself contain subordinate resonance structures, while simultaneously participating in larger resonance organizations. RQT therefore considers physical organization to be hierarchical: an identity may act as a coherent whole at one level while remaining internally composed of interacting sub-identities at another.

Changes introduced at one level need not remain confined to that level. An external resonance influence may alter a relationship within the larger identity, while the resulting mismatch is redistributed through its internal organization. Conversely, changes arising within a subordinate structure may propagate outward and affect the relationships through which the higher-level identity is maintained.

The response of a resonance system therefore depends on the distribution of its available accommodation capacity. Some disturbances may be absorbed locally, requiring only minor adjustment within a limited part of the structure. Others may be distributed across several internal relationships, producing a broader but still coherent reorganization. Where the existing organization cannot accommodate the imposed mismatch, the disturbance may extend to the system boundary and alter its external relationships.

This distribution is not assumed to follow a predetermined geometric pathway. It reflects the evolving compatibility of the resonance relationships participating in the system. Accommodation proceeds through those parts of the organization capable of contributing to a more persistent shared evolution, while strongly incompatible relationships may weaken, reorganize, or separate.

A higher-level identity may therefore remain stable even while its internal resonance organization changes substantially. Its persistence depends not on the immutability of its components, but on whether the combined adjustments preserve the relationships through which the whole remains distinguishable. In this sense, stability at one level may be supported by continuous reorganization at another.

Resonance structures are also constrained by their wider environment. Because each identity participates in several simultaneous relationships, no accommodation occurs in isolation. A configuration compatible with one relationship may introduce mismatch into another. Persistent organization therefore emerges from the distributed accommodation of multiple, mutually constraining resonance influences.

Where accommodation succeeds, previously distinct structures may participate in a more coherent higher-level identity. Where it fails, the existing organization may release subordinate

structures, alter its external relationships, or separate into new resonance systems. Formation and dissolution are thus complementary outcomes of the same underlying process: the redistribution of resonance toward configurations capable of continued organization.

This hierarchical view prepares the transition from general resonance processes to the structured relationships through which persistent resonance organization can be represented. The following sections introduce the resonance channels, closures, and enclosures that describe these enduring forms of shared resonance evolution.

4 Resonance Channels and Closure

The formation of observable structure.

The preceding sections described resonance organization as a continuing process of interaction, accommodation, redistribution, and persistence. As the number of participating structures and relationships increases, however, a purely process-based description becomes increasingly difficult to follow. A compact representation is therefore required for the persistent organization produced by these processes.

RQT introduces the concepts of resonance channels and resonance closure for this purpose. These concepts do not add new physical ingredients to the theory. A resonance channel represents the continuing organization of a persistent interaction, while closure describes the extent to which that interaction has become incorporated into a self-consistent resonance structure.

A channel is therefore not assumed to be a material conduit or a geometrically defined path. At the general level of RQT, it expresses a persistent resonance relationship whose organization contributes to the identities of the participating structures and, where applicable, to the higher-level identity they form together.

Not every resonance relationship becomes fully closed. Some remain partially accommodated, externally exposed, or distributed across wider resonance systems. The distinction between open, residual, and closed resonance organization will later allow increasingly complex systems to be described without requiring their complete internal evolution to be followed at every level.

These concepts remain independent of any particular microscopic geometry. The Roton Quantum Model will later propose specific structural representations of resonance channels, nodes, closures, and enclosures. At this stage, channels and closure are introduced only as the general relational language through which persistent resonance organization can be described.

4.1 Resonance Channels

A resonance channel is the persistent organization of a distinguishable resonance interaction.

Resonance interaction may be transient, diffuse, distributed, or continuously changing. Where such interaction becomes sufficiently persistent and organized to contribute to the continuing identity of a resonance structure or system, RQT represents it as a resonance channel.

A channel is therefore not an additional physical substance, conduit, or pre-existing path through space. It is the relational organization through which resonance structures participate in a shared evolution. The interaction is fundamental; the channel is its persistent and distinguishable form.

A resonance channel need not initially possess a defined geometric length, width, curvature, or orientation. At the general level of RQT, it is characterized by the persistence, compatibility, and organization of the interaction it represents. Specific geometric properties may arise only when the channel is embedded within a more detailed structural model.

Channels may differ in the way they participate in a resonance system. Some form persistent relationships between identifiable counterparts. Others remain externally exposed, distributed across several interactions, or only partially organized around a dominant partner. A channel

may therefore be open, relationally occupied, redirected, or incorporated into a more completely closed resonance organization.

The existence of a channel also does not require an unchanging interaction partner. What persists may be the organized mode of coupling itself, while the structures contributing to it change over time. This allows a channel to represent both stable pairwise relationships and persistent interactions with a distributed resonance environment.

A realized channel contributes to the identities of the structures it connects. Where several resonance structures become organized through persistent channels, their relationships may together establish a distinguishable higher-level resonance identity. Channels therefore do not merely connect pre-existing objects; they participate in defining what those objects and larger systems become.

The number and mutual compatibility of channels available to a given identity are not determined by general RQT alone. They depend on the adaptability and internal organization of the resonance structure supporting them. The Roton Quantum Model later represents these constraints through specific channel types, channel counts, and structural arrangements.

4.2 Open and Closed Channels

A realized resonance channel may remain open or become increasingly closed according to how fully its interaction is incorporated into the resonance organization under consideration.

An open resonance channel forms a persistent relationship between distinguishable resonance structures, but remains at least partly available to further accommodation. Its organization may still respond substantially to external resonance influences, shift between competing relationships, or participate in the formation of a larger resonance system. Openness therefore does not imply that the channel is weak or temporary. It indicates that the relationship has not yet become fully internalized within a stable resonance organization.

A closed resonance channel is a persistent interaction that has become predominantly incorporated into the internal organization of a resonance system. Closure does not terminate or deactivate the relationship. Rather, the participating structures sustain one another through an increasingly self-consistent resonance process, making the channel part of the identity of the resulting system.

The distinction between open and closed channels is not necessarily binary. Continued accommodation may gradually increase closure, while changing resonance conditions may partially or temporarily reopen a previously closed channel. During such reopening, the system becomes more accessible to external accommodation and may redistribute, redirect, or reorganize the relationship.

Whether a channel is considered open or closed also depends on the identity level being examined. A channel may be closed within one resonance system while the resulting higher-level identity remains open to additional relationships of its own. Closure at one level therefore does not imply isolation from the wider resonance environment.

By becoming closed, a channel changes more than its own relational state. Its incorporation alters the resonance organization of the participating structures and thereby changes which further relationships remain compatible and available to the resulting identity.

Closure changes the future compatibility landscape of the resulting identity.

4.3 Residual Resonance

Residual resonance is resonance influence that remains externally accessible without being fully organized into a closed relationship.

Residual resonance may arise where a realized channel is only incompletely closed, temporarily reopened, or continuously affected by redistribution within its enclosing structure. In such cases,

part of the interaction remains accessible to the wider resonance environment even though the principal relationship is sustained internally.

Residual resonance may also arise from an available channel mode that has not become organized around a distinct counterpart. Its influence then remains distributed across the surrounding resonance environment rather than forming a directed, partner-specific channel. Another compatible structure may respond to this distributed influence and, through continued accommodation, establish a realized resonance channel.

Residual resonance therefore differs from a closed channel both in organization and relational specificity. A closed channel sustains a distinguishable relationship within an enclosure, whereas residual resonance may remain diffuse, externally exposed, and shared among several possible interactions.

The strength and distribution of residual resonance may later be related to the emerging separation and geometry of resonance structures. Their quantitative representation, including possible distance-dependent laws, belongs to the more specific development of the Roton Quantum Model.

4.4 Resonance Closure

Resonance closure occurs when a persistent interaction becomes sufficiently accommodated to form an internally sustaining part of a resonance identity.

A realized resonance channel may remain externally exposed, compete with alternative relationships, or require continued adjustment. Through accommodation, however, it may become increasingly self-consistent and internalized within a larger resonance organization.

Closure does not terminate the interaction. It changes its relational role: an externally negotiable relationship becomes an internally supporting component of the resulting system. This transition may be gradual, and changing resonance conditions may partially reopen or reorganize a previously closed channel.

Closure also changes which further relationships remain compatible. As a channel becomes internalized, equivalent external coupling modes may be reduced, redirected, or removed. Depending on the identity, this may saturate a channel type or leave other compatible channels available.

Closure does not imply perfect isolation. Residual influence and weak environmental redistribution may remain. It denotes a state in which the relationship is predominantly sustained within the internal organization of the resonance system, thereby preparing the emergence of a higher-level identity.

4.5 Resonance Stabilization

Closure favours stability by converting externally negotiable resonance relationships into internally sustaining organization.

Reduced external coupling may arise through several distinct processes. A relationship may become internally closed through resonance locking, while rapid rotation or other continuing variation may prevent a stable counterpart from forming and leave only residual external influence.

Alternatively, interacting sub-identities may reorganize into a new combined resonance mode in which the original channel becomes unavailable and different internal or external channels emerge.

Closure, dynamical decoupling, and internal mode reorganization may therefore all stabilize a resonance system, but through different changes in its underlying organization.

4.6 Channel Availability and Saturation

A resonance identity can realize only those channels that remain compatible with its existing resonance organization.

Channel availability describes the forms of persistent interaction that an identity is capable of sustaining. An available channel mode is not yet a realized relationship; it becomes one only when compatible structures establish a sufficiently persistent interaction.

Realizing or closing a channel changes the organization of the identity and may preserve, reduce, redirect, or remove the compatibility required for additional channels. Several channels can coexist only where their accommodation demands remain mutually compatible.

An identity is saturated with respect to a channel type when no additional compatible channel of that type can be realized within its present organization. Residual interaction may nevertheless remain, and reopening, weakening, or reorganization of existing channels may later restore availability.

The number and arrangement of channels supported by particular physical identities are not determined by general RQT, but by the more specific structures proposed within the Roton Quantum Model.

Channel availability may be reduced by realized or closed relationships; where no further compatible channel of a given type remains, the identity is saturated with respect to that channel type.

4.7 Resonance Enclosure

A resonance enclosure is a persistent higher-level identity sustained by relationships that have become predominantly internal to its organization.

When resonance channels close, the participating structures do not lose their subordinate identities. Their resonance possibilities are instead constrained and reorganized by the combined system. The resulting enclosure can persist and interact as a distinguishable identity of its own.

An enclosure is not necessarily a spatial shell or an isolated system. Its boundary is defined by the distinction between relationships that sustain the identity internally and those through which it remains connected to the wider resonance environment.

Formation of an enclosure does not end accommodation. Continued internal redistribution may allow the system to enter more coherent resonance states and establish internal modes that were unavailable to its constituents separately. In doing so, previously external channel availability may be reduced or reorganized, while new relationships become possible within the enclosure.

Different enclosures may internalize different channel types or different combinations of relationships. Their stability therefore depends not only on the degree of closure, but also on whether the resulting internal organization can continue to accommodate external resonance demands.

An enclosure may itself retain open channels, residual resonance, and further channel availability. It can therefore participate as a sub-identity in a larger enclosure, extending resonance organization across successive hierarchical levels.

As accommodation progresses, one shared resonance organization may become increasingly dominant over competing configurations. Further adjustments then reinforce the same relationship, dynamically confining the interaction to an increasingly compatible mode.

Locking may require substantial reorganization within the participating identities. Existing channels may shift or temporarily reopen, while resonance that cannot be incorporated into the emerging closure may be redistributed or released.

The process need not be instantaneous or irreversible. Locking may remain incomplete, become temporarily stable, or reverse under changing resonance conditions. Where it succeeds, the realized channel becomes increasingly internalized and may culminate in channel closure and the formation or reorganization of a resonance enclosure.

Locked and closed are therefore not fully synonymous. Closure describes how completely a relationship has become incorporated into the internal organization of an enclosure. Locking describes the dynamical capture of that relationship and its resistance to leaving the established resonance mode. A channel may consequently be strongly closed yet comparatively easy to reopen, or strongly locked while closure remains incomplete.

Closure describes the resulting relational structure; locking describes its dynamical capture and retention.

5 Relational Separation and Emergent Geometry

The preceding sections described how persistent resonance relationships may close and become internal to a higher-level resonance enclosure. Closure, however, is not the only way in which several resonance identities can form a stable organization.

Distinct identities may remain individually recognizable while participating in a shared resonance structure at the same organizational level. Additional identities may join this structure without necessarily becoming subordinate components of a recursively enclosed identity. Their relationships instead form a lateral resonance network whose stability depends on the mutual compatibility of the complete arrangement.

Such networks introduce relational separation. The participating identities must remain sufficiently distinct to preserve their individual organization, yet sufficiently compatible to sustain persistent resonance relationships. Separation is therefore not initially understood as distance within a pre-existing metric space. It describes the relational distinction required for several identities to coexist within a common resonance organization.

5.1 Relational Position

A resonance identity acquires a relational position through the pattern of interactions in which it participates. Its position is initially defined not by coordinates, but by its relationships to neighbouring identities and by the role it occupies within the wider resonance organization.

Two otherwise equivalent identities may therefore occupy different relational positions because they participate in different channel configurations. As these configurations become persistent and repeatable, they establish an ordered distinction between the identities involved.

Relational position provides the first basis for later concepts such as neighbourhood, direction, and location. These concepts do not precede resonance organization, but arise from persistent differences within it.

5.2 Resonance Separation

Resonance separation describes the relational distinction maintained between identities that participate in a shared organization. It expresses the degree and form of differentiation required for their individual resonance structures to remain distinguishable while continuing to interact.

This separation need not initially be metric or uniquely defined. The same pair of identities may be closely related through one channel type while remaining more weakly connected through another. A geometric notion of distance becomes meaningful only where resonance relationships establish repeatable and mutually comparable separations.

Separation may also contribute to stability. Greater proximity does not necessarily improve accommodation. Where nearby resonance structures impose mutually incompatible demands, a larger separation may reduce the unresolved interaction and permit a more persistent combined organization.

5.3 Phase Compatibility and Propagation

Structural similarity alone does not guarantee that resonance identities can sustain a shared relationship. Their recurring evolution must also meet in compatible relative states. RQT refers to this relational condition as phase compatibility.

Phase is not introduced as an absolute temporal coordinate. It describes where one resonance structure stands within its recurring evolution relative to another. Two otherwise identical identities may therefore require a particular phase relation in order to sustain a persistent channel.

Where a resonance relationship extends across relational separation, the influence of one identity cannot be considered independently of how that influence continues through the relationship. Propagation describes this continued transfer of resonance organization across a relationally extended interaction.

Phase compatibility may consequently depend on both the recurring evolution of the participating identities and the separation over which their resonance influence is maintained. Several combinations of phase and separation may support a stable shared mode; compatibility need not select one unique relation.

5.4 Distance Locking

Where a particular combination of phase relation and separation repeatedly supports compatible resonance evolution, the participating identities may stabilize that relation. RQT describes this condition as distance locking.

Distance locking differs from resonance closure. Closure internalizes a relationship within a higher-level enclosure, whereas distance locking preserves a stable separation between identities that remain distinguishable participants of the same lateral network.

A locked separation need not represent a rigid or exact distance. The relationship may tolerate a range of variation, and surrounding resonance demands may shift the preferred relation. Stability lies in the continued recurrence of compatible organization rather than in an immutable geometric interval.

5.5 Lateral Resonance Networks

When several identities establish mutually compatible channels at the same organizational level, their relationships form a lateral resonance network. In principle, additional identities may join such a structure without changing its basic organizational span, provided that the resulting resonance demands remain jointly compatible.

The stability of the network cannot generally be reduced to isolated pairwise relations. Each identity may need to maintain compatible phase and separation relations with several neighbours simultaneously. Adjusting one relationship may therefore improve or disturb others.

This produces resonance frustration where no arrangement can satisfy all preferred relations at once. The network may respond by shifting phase relations, modifying separations, selecting among possible channels, reorganizing its neighbourhood structure, or separating into more compatible clusters.

Stable network geometry therefore represents an accommodated compromise among several preferred resonance relations rather than the exact repetition of one ideal pairwise distance.

5.6 Emergent Geometry

Persistent relational positions, repeatable separations, and compatible phase arrangements provide the basis for geometric description. Where several identities sustain distinct relationships simultaneously, their organization may acquire direction, angular relation, neighbourhood, and dimensional structure.

Geometry is therefore introduced as a representation of stable resonance relations rather than as a pre-existing container in which resonance structures are placed. A geometric distance represents a reproducible relational separation, while direction and dimension express distinguishable ways in which several such relations can coexist.

The dimensionality and metric properties of the resulting organization are not fixed by general RQT alone. They depend on the number and compatibility of relationships that particular resonance identities can sustain, together with the characteristic phase and propagation conditions of those relationships.

The Roton Quantum Model later represents these principles through nodes, channel arrangements, characteristic lengths, and concrete resonance networks.

5.7 Dimensional Selection

The emergence of geometric organization does not by itself determine why persistent physical networks are expressed predominantly through three spatial dimensions. RQT allows this observed dimensionality to be interpreted as a stability condition of externally connected resonance systems rather than as a primitive assumption.

Additional relational dimensions would provide further possibilities for organization, but would also increase the number of channel relations, phase conditions, and accommodation demands that must remain mutually compatible. Three-dimensional networks may therefore represent a particularly stable compromise between relational flexibility and the complexity required to preserve coherent organization.

This does not exclude richer internal resonance structures. Additional degrees of freedom may remain confined within subordinate enclosures, recurring modes, or internally closed channel arrangements without appearing as independent external spatial dimensions. An identity may consequently possess a more complex internal organization while participating in its environment through a predominantly three-dimensional resonance geometry.

Once such an external geometry has become persistent, changing it requires coordinated redistribution across the relationships that sustain the whole. Dimensional organization therefore prepares the concept of inertia as resistance to the reconfiguration of an established resonance network.

6 Information, Persistence and Inertia

A persistent resonance identity does not exist as a sequence of independently recreated states. Its present organization continues from its previous organization and thereby carries the constraints of its own history. Phase relations, channel configurations, closures, and ongoing transitions together determine which further developments remain compatible with the identity.

RQT therefore does not treat information as an additional substance stored within a resonance structure. Information is expressed by the organization that persists and by the alternatives that this organization permits or excludes. The state of a resonance system is informative because it influences how the system can continue.

6.1 Information as Persistent Organization

A resonance identity preserves information through the continuity of its internal and relational organization. Its present state reflects earlier accommodation processes and cannot generally be changed without affecting the relationships through which the identity remains coherent.

This information need not correspond to a static record. It may be carried by recurring phase relations, ongoing transitions, closed channels, or stable patterns of redistribution. A resonance structure therefore retains its history dynamically: the past remains effective through the organization that continues into the present.

Persistence does not require every internal state to remain unchanged. Subordinate relationships may oscillate, reopen, redistribute, or reorganize while the higher-level identity remains distinguishable. What persists is the continuity of the organization that connects these changing states.

6.2 Resistance to Reconfiguration

Because a persistent resonance system is relationally organized, a change imposed upon one part cannot always remain local. Altering a phase relation, channel state, or internal configuration may change the compatibility demands placed upon other parts of the structure.

The system must then redistribute the resulting resonance mismatch until a new coherent organization is established. Where this redistribution involves many mutually constrained relationships, the response cannot occur as an independent and instantaneous change of a single component.

RQT identifies the resulting resistance or lag as inertia. Inertia is therefore not initially defined as resistance to motion through space, but as resistance to the reconfiguration of an established resonance organization.

The strength of the inertial response depends on how extensively the imposed change must be accommodated. A local adjustment that remains compatible with the surrounding structure may produce little resistance. A change that requires coordinated reorganization across many channels, closures, or subordinate identities produces a greater system-wide response.

Inertia does not prevent change. It expresses the continuity that must be preserved while change proceeds. The existing organization constrains the path by which the system can reach another coherent state.

6.3 Inertia Across Resonance Hierarchies

A resonance identity may contain subordinate structures while participating in larger resonance organizations. Reconfiguration can therefore be distributed across several hierarchical levels.

An imposed change may be absorbed within a local relationship, redistributed through the internal organization of an enclosure, or transferred to the higher-level identity as a whole. Which response occurs depends on the available accommodation modes and on the stability of the relationships involved.

A higher-level identity may remain persistent while substantial reorganization occurs internally. Conversely, a disturbance that cannot be accommodated within subordinate structures may alter the organization of the entire enclosure or propagate into its external relationships.

Inertia is therefore relative to the identity level under consideration. What appears as internal adaptation at one level may appear as delayed response, reorientation, or structural change at another.

The persistence of a resonance identity and its inertia are thus closely connected. Persistence carries an established organization forward; inertia expresses the resistance encountered when that organization must be changed.

A locally imposed phase change may therefore spread through the resonance organization as a sequence of accommodation demands rather than being adopted simultaneously by the entire structure.

7 Hierarchical Resonance Structures

Persistent resonance organization may be represented at different levels of structural detail. A higher-level identity can be treated as a coherent unit while its internal resonance organization remains hidden, or it can be expanded into the subordinate identities and channels through which it is sustained.

This hierarchical representation does not introduce additional physical entities. It provides a way to describe the same resonance organization at different levels of resolution. At each level, internally coherent structures may be represented as nodes, while their persistent same-level relationships are represented as resonance channels.

7.1 Resonance Nodes

A resonance node is a distinguishable resonance identity represented at a chosen organizational level by the channels through which it participates in a wider resonance network.

A node is not necessarily elementary and should not initially be understood as a geometric point. It may represent a minimal resonance identity, a resonance enclosure containing subordinate structures, or a larger coherent system whose internal organization is not relevant at the level currently examined.

The node representation compresses this internal organization while preserving the resonance properties required for its external relationships. These may include its present phase state, available and realized channel modes, closure conditions, and compatibility with neighbouring nodes.

Channels connected to a node are not attached to a passive junction. The node accommodates the relational conditions imposed by its incident channels, and its own resonance state constrains which relationships can remain compatible. A change in one relationship may therefore alter the accommodation demands placed upon the others.

Nodehood is relative to the chosen level of description. A structure represented as one node in a higher-level network may itself be expanded into an internal network of subordinate nodes and channels. Conversely, a persistent lower-level network may be represented as one higher-level node once its combined organization becomes sufficiently coherent to participate externally as a unit.

Links between hierarchy levels should not be interpreted as additional resonance channels. They express that a higher-level node is realized by, and remains consistent with, a subordinate resonance network. Changes in the higher-level identity therefore correspond to redistributed phase or resonance conditions within the lower-level organization, without requiring a separate cross-level channel.

The Roton Quantum Model later assigns concrete internal structures, channel types, and geometric representations to particular nodes. At the general level of RQT, the node remains a scale-dependent representation of persistent resonance identity.

7.2 Resonance Networks

7.3 Resonance Clusters

7.4 Emergent Hierarchies

8 From Relational Resonance to the Roton Quantum Model

8.1 From Relational Position to Directionality

8.2 From Resonance Separation to Geometric Distance

8.3 From Channels to Spatial Representation

8.4 Model Scope and Additional Assumptions

8.5 From Relational Resonance to Directionality

RQT does not begin with space, direction, or dimensionality. It begins with identities and resonance relationships. A resonance channel is therefore not yet a line in space, but a relational

constraint between identities. Distance, direction, and geometric alignment only become meaningful once such relational constraints are realized as a persistent structure. This is the conceptual bridge from the resonance theory (RQT) to a concrete model (RQM).

A whole nucleus could, in principle, be described as a pure resonance graph without assigning any three-dimensional position to its nodes. Geometry only enters when such a graph is projected into a spatial model. This raises a natural question: if stable resonance graphs are later mapped into space, which dimensionality would they prefer? Would the most stable realization be three-dimensional, four-dimensional, or even higher-dimensional? RQT itself does not postulate the answer. It only provides the relational structure.

The first important observation is that pure phase accommodation on a graph does not by itself select a preferred dimensionality. Simple simulations of resonance networks with identity nodes, channel nodes, local phase states, and inertial phase accommodation show that many symmetric topologies can settle into coherent oscillatory states. This suggests that dimensionality is not selected by topology alone. A graph can distribute phase, but it does not yet know what it means for two channels to be parallel, opposite, or linearly aligned.

The missing concept is not dimensionality itself, but directionality. In RQM, directionality appears through axial rotations and the alignment of resonance projections along shared axes. This axial alignment was previously postulated as part of the 3D model. From the RQT perspective, however, such alignment can be motivated by coherent propagation through the relational graph. A graph may possess entry and exit nodes such that resonance entering at one location can split into several equivalent paths, propagate through the structure, and recombine at a distinct opposite location. In that case, the graph has a directional propagation property before any geometric direction has been assigned.

This changes the interpretation of linear alignment. Linearity is not fundamental in RQT. Instead, it is the RQM realization of coherent relational propagation. When a resonance can pass through a structure and reappear at a distinguished opposite node with minimal internal accommodation, the later geometric embedding naturally represents this as an effective direction. Thus, what appears in RQM as a straight resonance axis is, at the RQT level, a stable entry–propagation–recombination path through the graph.

This also clarifies why some graph structures are nucleus-relevant while others are not. A proton-like isolated structure (e.g. tetrahedron) may expose residual channels, but it does not yet provide a strong global entry–exit propagation axis. A neutron-like isolated structure (e.g. triangular dipyramid) may have one weak linear possibility, but lacks stable topological continuation as an isolated object. A deuteron-like structure (e.g. octahedron) already offers several propagation axes, while an alpha-like structure has a much richer set of coherent propagation possibilities. In an icosahedral interpretation of the alpha particle, there are naturally six antipodal directions through which resonance can enter, circulate through multiple paths (around the sphere), and recombine. This gives the alpha particle a strong resonance potential in several directions.

The emergence of 3D can then be understood as an economy of resonance realization. Higher-dimensional regular structures, such as the 16-cell or 24-cell, provide elegant symmetry extensions. However, higher symmetry alone is not sufficient. A structure is favourable only if it increases coherent internal propagation while reducing exposed residual dependency. If a 4D extension adds many additional vertices and channels without improving the charge–mass–residual-channel economy, then combinations of 3D structures may be more efficient. In this sense, nature need not maximize dimensionality. It may instead maximize reusable coherent resonance per exposed dependency.

RQM is therefore not introduced as an arbitrary 3D assumption, but as the dominant geometric realization of resonance structures whose relational propagation already contains effective directions. The RQT graph supplies coherence paths; RQM realizes them as axial resonance channels in 3D. Linear resonance alignment is then not merely a geometric convenience, but the visible projection of stable relational propagation. This provides a foundation for the central RQM idea

that normal matter is built from aligned rotational resonance structures in three-dimensional space.

9 Local Resonance Optimization

9.1 What Does a Resonance System Optimize?

Standard physical descriptions commonly characterize stable systems as configurations of minimal energy. A transition toward a more stable state is therefore described as a descent toward a lower energy level, accompanied by the release of excess energy.

Within Resonance Quantum Theory, this description is inverted. A stable system is not regarded as a system that has merely lost energy. It is a system that has found a configuration in which a larger portion of its available resonance capability can be maintained within persistent, mutually compatible resonance relations.

The local optimization problem is therefore not primarily the minimization of an abstract energy quantity. It is the search for a reachable configuration in which the remaining externally exposed resonance capacity is reduced relative to the resonance capability maintained by the system.

A resonance structure is locally improved when it can:

- close more of its internal resonance channels,
- reduce the strength or number of externally exposed residual channels,
- distribute unavoidable mismatch across larger relational distances,
- maintain its resonance structure over longer temporal intervals,
- or separate a poorly accommodated substructure from the remaining enclosure.

The locally preferred state is therefore not necessarily the globally most enclosed state. A system can only enter configurations that are dynamically reachable through its presently available resonance channels.

This distinction is important for metastable structures and isotopes. An isotope need not be intrinsically unstable in an absolute sense. It may persist indefinitely as long as no nearby and dynamically accessible configuration offers a substantially improved resonance enclosure.

A decay process becomes possible when the existing structure encounters a pathway toward a configuration with lower residual exposure. Such a pathway may depend on internal oscillations, environmental coupling, nearby identities, or temporary resonance mismatches.

The optimization principle may therefore be stated as follows:

Local resonance dynamics favor reachable configurations that maintain the greatest resonance capability while exposing the least residual channel capacity to their surroundings.

9.2 Resonance Capability and Residual Exposure

Let a resonance enclosure possess a total effective resonance capability

$$\mathcal{R}, \tag{1}$$

representing the amount of resonance activity that can be maintained, transferred, circulated, or committed to resonance channels.

This quantity should not yet be identified directly with conventional energy. It represents the ability of a structure to sustain resonance relations and to participate in further resonance formation.

A portion of this capability is committed to internally maintained resonance structures. We denote this part by

$$\mathcal{R}_{\text{enc}}, \quad (2)$$

whereas the externally exposed channel capacity is denoted by

$$\mathcal{C}_{\text{res}}. \quad (3)$$

The residual capacity includes all resonance channels that remain externally accessible, insufficiently matched, only temporarily answered, or dependent on the surrounding resonance landscape.

A first non-normalized measure of enclosure quality is then

$$\mathcal{Q}_{\text{enc}} = \frac{\mathcal{R}_{\text{enc}}}{\mathcal{C}_{\text{res}}}. \quad (4)$$

The quantity $\mathcal{Q}_{\text{enc}}$ may be called the *Resonance Enclosure Quotient*.

A structure improves its local resonance condition when

$$\mathcal{Q}_{\text{enc}} \rightarrow \text{maximum}. \quad (5)$$

The quotient is deliberately not normalized. A normalized ratio would conceal an important limiting behavior. If the residual channel capacity approached zero,

$$\mathcal{C}_{\text{res}} \rightarrow 0, \quad (6)$$

then

$$\mathcal{Q}_{\text{enc}} \rightarrow \infty. \quad (7)$$

Complete closure on all resonance spans would therefore correspond to an infinite limiting value rather than an ordinary physical state.

This suggests that absolute resonance closure is not a realizable configuration. A structure may strongly reduce its residual channels, but it cannot remove its resonance capability from the relational universe altogether.

A fully closed structure would possess no remaining channel through which it could:

- interact,
- propagate,
- be distinguished,
- be localized,
- or participate in a higher-level resonance identity.

Absolute closure would therefore remove the relational conditions under which the structure could remain part of physical reality.

This leads to a possible RQT interpretation of energy conservation:

Resonance capability may be redistributed, enclosed, propagated, or transferred, but it cannot be removed from relational availability altogether.

9.3 Resonance-Residual Ratio

For comparisons between structures, a complementary quantity may be useful. The *Resonance-Residual Ratio* describes the residual channel capacity relative to the resonance capability maintained by the enclosure:

$$\text{RRR} = \frac{\mathcal{C}_{\text{res}}}{\mathcal{R}_{\text{enc}}}. \quad (8)$$

The Resonance-Residual Ratio is the inverse of the Resonance Enclosure Quotient:

$$\text{RRR} = \frac{1}{\mathcal{Q}_{\text{enc}}}. \quad (9)$$

A low RRR indicates that a large amount of resonance capability is maintained with comparatively little external exposure. A high RRR indicates that the structure remains strongly dependent on external coupling or carries substantial unresolved resonance mismatch.

The optimization principle may therefore be written equivalently as

$$\mathcal{Q}_{\text{enc}} \rightarrow \text{maximum}, \quad (10)$$

or

$$\text{RRR} \rightarrow \text{minimum}. \quad (11)$$

The positive formulation is conceptually preferred:

Nature locally maximizes resonance enclosure.

The minimization of residual exposure is then a consequence rather than the primary description.

9.4 Environmental Dependence of Channel Closure

Whether a channel is open, residual, reactive, or effectively closed cannot generally be determined from the subsystem alone.

A potential channel may remain unanswered in one environment and become fully accommodated in another. An externally extending channel is not necessarily residual if it participates in a stable higher-level resonance enclosure.

Accordingly, the effective residual capacity should be written as a function of both the subsystem S and its resonance environment $\mathcal{E}$:

$$\mathcal{C}_{\text{res}} = \mathcal{C}_{\text{res}}(S, \mathcal{E}). \quad (12)$$

The enclosure quotient is therefore also relational:

$$\mathcal{Q}_{\text{enc}} = \mathcal{Q}_{\text{enc}}(S, \mathcal{E}). \quad (13)$$

The same internal structure may consequently be stable in one environment and unstable in another.

This provides a natural interpretation of environmental decay, catalysis, induced transitions, and resonance competition. The surrounding structure does not merely add external energy. It changes which resonance channels can be answered and which alternative enclosures become reachable.

A channel may be considered effectively accommodated when it is:

1. internally closed within the subsystem,

2. persistently coupled to a compatible external identity,
3. integrated into a stable higher-level enclosure,
4. or maintained through a temporally coherent oscillatory relation.

A channel remains residual when no sufficiently compatible continuation is available within the relevant resonance distance and time.

9.5 Decay as Resonance Reorganization

Consider an initial resonance structure S that transforms into a remaining enclosure E and an emitted structure X :

$$S \rightarrow E + X. \quad (14)$$

In conventional language, the system releases energy and falls into a lower-energy state. In RQT, the process is interpreted as a redistribution of resonance capability:

$$\mathcal{R}_S = \mathcal{R}_E + \mathcal{R}_X, \quad (15)$$

subject to the appropriate resonance relations between the resulting structures.

The transformation becomes locally favorable when the remaining structure achieves an improved enclosure quotient:

$$\mathcal{Q}_{\text{enc}}(E) > \mathcal{Q}_{\text{enc}}(S). \quad (16)$$

The emitted structure carries away resonance capability that could not be maintained efficiently within the previous enclosure. Its emission allows the remaining resonance pattern to reduce mismatch and to close a greater portion of its available channels.

The system has not become stable merely because it contains less conventional energy. It has become stable because the remaining resonance capability is organized more effectively.

This may be summarized as follows:

A resonance enclosure may discard part of its structure when separation allows the remaining resonance capability to achieve a more complete local enclosure.

Matter–antimatter annihilation illustrates the general character of this principle. The material enclosures cease to exist, but their resonance capability does not vanish. It reappears in propagating radiative structures.

The resulting radiation demonstrates that even a seemingly complete material cancellation does not produce absolute closure. The resonance capability remains relationally active.

9.6 Propagation as Temporal Channel Closure

The enclosure principle also applies to propagation.

A freely propagating structure reduces its effective exposure to a local environment by shortening the interval during which its residual channels remain accessible at any one location.

The effective residual capacity may therefore depend not only on structural channel strength, but also on local exposure time:

$$\mathcal{C}_{\text{res}}^{\text{eff}} \sim \mathcal{C}_{\text{struct}} \tau_{\text{exp}}, \quad (17)$$

where $\mathcal{C}_{\text{struct}}$ represents the intrinsic residual coupling capability and τ_{exp} the duration of local accessibility.

A propagating structure can improve its enclosure condition by reducing

$$\tau_{\text{exp}}. \quad (18)$$

Propagation is therefore not merely displacement through an already existing space. It can serve as a form of *temporal channel closure*.

A freely propagating photon may represent an extreme realization of this principle. Its internal resonance structure achieves its strongest enclosure when it propagates at the maximal coherent Sonon transition rate.

The photon does not eliminate all external channels. Otherwise, it could never be emitted, absorbed, or scattered. Instead, it reduces their local temporal accessibility.

A photon interacts again when a sufficiently compatible material or radiative resonance structure occupies the same local resonance region and can open a new channel configuration.

The transition may be represented schematically as

$$\text{free propagative enclosure} \rightarrow \text{temporary channel opening} \rightarrow \text{coupled transition structure} \rightarrow \begin{cases} \text{absorption,} \\ \text{scattering,} \\ \text{re-emission.} \end{cases} \quad (19)$$

During absorption, the free photon identity does not first become a slower photon. Its propagative enclosure opens and its resonance capability becomes integrated into the receiving system.

9.7 The Sonon Origin of the Local Propagation Limit

Within the Roton Quantum Model, the maximal propagation speed is not introduced as an abstract prohibition. It is associated with the geometry and period of the fundamental Sonon process.

Let

$$T_s \quad (20)$$

denote the characteristic Sonon period, and let

$$\ell_s \quad (21)$$

denote the maximal coherent transition length that a Sonon-based resonance structure can bridge during one period.

The maximal coherent propagation speed is then

$$c_s = \frac{\ell_s}{T_s}. \quad (22)$$

In the simplest geometric interpretation, ℓ_s may be related to the characteristic Sonon diameter:

$$\ell_s \approx d_s. \quad (23)$$

This yields

$$c_s \approx \frac{d_s}{T_s}. \quad (24)$$

The relevant length is not necessarily the complete rotational circumference. It is the maximal displacement over which the resonance structure can maintain cyclic self-closure between successive states.

For a propagating Sonon structure with velocity v , temporal closure requires

$$vT_s \leq \ell_s. \quad (25)$$

If

$$vT_s > \ell_s, \quad (26)$$

the next state lies beyond the coherent continuation range of the previous state. The resonance structure can no longer maintain its identity and must reorganize or disintegrate.

The local speed c observed in the measurable universe may therefore correspond to

$$c = c_s. \quad (27)$$

In this interpretation, c is the maximal coherent propagation speed of Sonon-based identities.

It appears universal because all presently measurable matter and radiation either consist of Sonon structures or interact through Sonon-compatible resonance channels.

The statement is therefore not necessarily that no imaginable structure could propagate faster than c . Rather:

No freely propagating Sonon-based resonance identity can exceed the maximal transition rate at which its cyclic structure remains self-consistent.

9.8 Why Not Every Sonon-Based Structure Moves at c

The fact that the Sonon process defines a maximal transition rate does not imply that every Sonon-based identity must propagate at that rate.

A resonance enclosure may use its available resonance capability for different forms of closure:

- axial propagation,
- internal rotation,
- orthogonal oscillation,
- standing resonance,
- coupling to neighboring structures,
- or integration into a higher-level identity.

For a photon, maximal axial propagation may offer the most effective enclosure. Its freely propagating structure therefore assumes

$$v_\gamma = c. \quad (28)$$

For an electron or another inertial enclosure, a substantial part of the resonance capability is maintained within internal circulation and persistent external channel structures. Its center of identity is therefore not required to propagate at c .

A neutrino-like structure may contain an additional weak orthogonal oscillation. If the total Sonon transition capability remains bounded by c , one may write schematically

$$v_{\parallel}^2 + v_{\perp}^2 = c^2. \quad (29)$$

Its forward propagation speed would then be

$$v_{\parallel} = \sqrt{c^2 - v_{\perp}^2}, \quad (30)$$

and therefore slightly below c .

The additional oscillatory structure would simultaneously create a more complex resonance footprint, reducing its compatibility with ordinary material resonance channels.

This remains a conceptual RQM interpretation and requires further geometric development.

9.9 Resonance-Distance Landscape

RQT does not begin with geometric distance as a fundamental quantity. Distance must emerge from the difficulty of establishing or maintaining resonance relations between identities.

The relational distance between two structures A and B may be represented by the minimal cumulative mismatch required to connect them through an admissible resonance path:

$$d_R(A, B) = \min_{\mathcal{P}: A \rightarrow B} \sum_{e \in \mathcal{P}} \mu_e, \quad (31)$$

where

- $\mathcal{P}$ is a possible sequence of resonance transitions,
- e denotes one transition within this sequence,
- and μ_e is the mismatch or transition cost associated with that step.

The apparently empty universe is therefore not devoid of relational structure. It forms a low-density resonance landscape shaped by the residual channels of all existing identities.

Large accumulations of matter contain vast internal resonance capability and correspondingly large distributions of residual mismatch. This mismatch must be accommodated through relational distance and through the surrounding transition landscape.

At this stage, no gravitational interpretation is required. The relevant claim is more primitive:

Concentrated resonance enclosures alter the surrounding distribution of resonance compatibility and resonance-distance requirements.

A propagating photon must continuously fit its resonance structure into this existing distance landscape.

It does not require a permanently closed long-range channel to a predetermined destination. Instead, it maintains its identity by realizing a sequence of locally compatible transitions.

The photon creates a propagative resonance line through the continued preservation of:

- frequency,
- phase relation,
- polarization,
- propagation orientation,
- and temporal closure.

A possible receiving structure becomes relevant only when its local resonance channels provide a more favorable continuation than free propagation.

9.10 Why Photon Propagation Remains Three-Dimensional

A photon originates from a material resonance structure with a three-dimensional extension and orientation.

Its initial resonance configuration therefore inherits compatibility with the shared three-dimensional resonance landscape established by material identities.

Higher-dimensional internal excursions are not necessarily prohibited. A resonance structure may locally use additional dimensions for:

- compactification,

- orthogonal oscillation,
- temporary channel rearrangement,
- or internal rotational closure.

However, long-range propagation requires continued compatibility with the surrounding resonance-distance landscape.

A persistent drift into an unsupported higher-dimensional direction would accumulate mismatch because the surrounding long-range resonance structure does not provide an equally continuous set of compatible transitions.

The propagating identity therefore tends to return to the dominant three-dimensional continuation.

This can be expressed conceptually as

$$\text{long-range propagation} \rightarrow \text{maximum compatibility with the shared 3D resonance landscape.} \quad (32)$$

Three-dimensional propagation is thus not imposed as an external geometrical rule. It emerges as the most broadly supported long-range resonance continuation.

Lower-dimensional internal structures may remain preserved within the propagating identity. Rotational planes, phase surfaces, and polarization structures can retain two-dimensional characteristics while the complete photon remains embedded in the shared three-dimensional landscape.

This provides a conceptual bridge between RQT and the geometric structures introduced by RQM.

9.11 Nonlocal Resonance Without Direct Signal Transfer

Consider a higher-level identity

$$C = A + B, \quad (33)$$

formed by a resonance relation between two spatially separated substructures A and B .

The identity C is not merely a communication line between two otherwise independent objects. It is a single higher-level resonance enclosure whose local manifestations appear at A and B .

The relation can dissolve only when both sides can enter alternative resonance configurations. Schematically,

$$C(A, B) \rightarrow C'(A', B'). \quad (34)$$

At A , the transition appears as a local change in environmental compatibility. At B , it likewise appears as a local change in environmental compatibility.

Neither side must observe a signal arriving from the other. Each side only observes that the previous higher-level identity can no longer be maintained and that a new local continuation has become available.

The transition is therefore locally causal at both ends while remaining globally constrained by the existence of the higher-level identity.

The nonlocal relation does not necessarily transmit freely selectable information. Instead, it restricts which local continuations can be jointly realized.

A measurable residual effect may appear in the switching inertia of the local structures. The time required for B to abandon its former channel and couple to a nearby structure B' may depend on whether the complete identity C remains capable of preserving its previous enclosure.

This could reveal information about the persistence or adaptability of the shared resonance relation, but not necessarily a freely encoded message sent from A .

The distinction may be summarized as follows:

A nonlocal resonance channel does not primarily transmit information between its endpoints. It defines the temporal and structural conditions under which both endpoints may continue to participate in the same identity.

9.12 Temporal Extension and Bi-Temporal Compatibility

No resonance exists at a dimensionless instant.

A realized resonance must persist across a finite temporal interval and maintain compatibility between preceding, present, and possible subsequent states.

A resonance identity should therefore not be represented solely as

$$R(t), \tag{35}$$

but more generally as a temporally extended structure

$$R[t_0, t_1]. \tag{36}$$

Its existence requires compatibility across both relational distance and relational time.

This does not imply that future events rigidly predetermine the present. The future acts not as a fixed destination, but as a constraint on which present resonance structures possess viable continuations.

The concept may be described as *pre-possibility constraint* rather than predestination.

A present resonance configuration is realizable only if it can be continued into at least one temporally compatible resonance history.

In this sense, the complete RQT description is bi-temporal:

$$R(t - \Delta t) \leftrightarrow R(t) \leftrightarrow R(t + \Delta t). \tag{37}$$

Forward causality remains the practical local description of physical change. Bi-temporal compatibility expresses the deeper requirement that a resonance identity must remain coherent across its complete temporal extension.

The principle may be stated as follows:

No resonance exists only at one place or one instant. Every realized resonance extends across resonance-compatible distance and resonance-compatible time.

9.13 Summary

Local Resonance Optimization replaces the image of physical systems descending toward abstract lower-energy states with the image of resonance structures searching for improved enclosure.

A locally favorable configuration maintains substantial resonance capability while reducing the capacity of externally exposed channels.

The central non-normalized quantities are

$$\mathcal{Q}_{\text{enc}} = \frac{\mathcal{R}_{\text{enc}}}{\mathcal{C}_{\text{res}}}, \tag{38}$$

and

$$\text{RRR} = \frac{\mathcal{C}_{\text{res}}}{\mathcal{R}_{\text{enc}}}. \tag{39}$$

The local dynamics tend toward

$$\mathcal{Q}_{\text{enc}} \rightarrow \text{maximum}, \tag{40}$$

or equivalently,

$$\text{RRR} \rightarrow \text{minimum.} \quad (41)$$

Complete closure is an unreachable limit because it would remove a structure from all relational participation.

Propagation can itself improve enclosure by reducing temporal exposure. For photons, maximal coherent propagation may therefore represent a form of temporal channel closure.

The measurable speed c is interpreted as the maximal coherent Sonon transition rate,

$$c = \frac{\ell_s}{T_s}, \quad (42)$$

rather than as an unexplained external restriction.

Distance emerges from resonance mismatch and transition cost. The apparently empty universe forms a resonance-distance landscape through which propagating identities must continuously maintain compatible closure.

Three-dimensional propagation remains dominant because long-range resonance continuation is most strongly supported within the shared three-dimensional structure inherited from material identities.

Finally, nonlocal resonance relations need not constitute direct information channels. They define higher-level identities whose local continuations must remain mutually compatible across both relational distance and temporal extension.

The combined principle is:

Physical structures persist by maximizing the enclosure of their resonance capability across compatible distance and compatible time.

10 The Roton Quantum Model

Discrete representation of resonance structures.

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10.2	Rotons
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11	Elementary Resonance Structures
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15.4	Numerical Simulation Approaches ³²
16	Predictions and Open Questions

18 Conclusions

Summary and outlook.

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The author is an independent researcher with a long-standing interest in foundational aspects of quantum theory, field dynamics and emergent phenomena in complex systems.

Companion Website and Further Material

A non-technical introduction, extended explanations, graphical illustrations and ongoing discussions of the Roton Quantum Model are available on the project website:

- Roton Quantum Model – intuitive overview and guided tour: <https://ledoigt.ch>
- Technical notes, simulations and updates: <https://ledoigt.ch/ledoigt/notes>
- Discussion and feedback channel: <https://ledoigt.ch/ledoigt/discuss>

These resources are intended as a more accessible companion to the present article, offering step-by-step motivation, visualizations, and evolving refinements of the model.

Data and Code Availability

Simulation scripts, illustrative numerical experiments and example configurations of Rotons used to explore the qualitative behavior reported in this work are (or will be) made available at:

- Code repository (numerical experiments): <https://github.com/ledoigt>
- Supplementary animations and figures: <https://ledoigt.ch/ledoigt/collections/>

Interested readers are invited to reproduce, extend and critically test the presented ideas.

Author’s Note

This work is intended as an open invitation to the theoretical physics and complex-systems communities to examine, challenge and refine the Roton Quantum Model. The author explicitly welcomes critical assessments, alternative formulations and concrete proposals for experimental or numerical tests, which may help to clarify the model’s scope, limits and possible connections to established frameworks.